

BAM 325: PRODUCTION MANAGEMENT

CONTROL CONCEPT IN PRODUCTION

Generally, control entails checking current performance against predetermined standards and targets contained in the plans, with a view to ensuring adequate progress and satisfactory performance.

Control is accomplished by monitoring the output, comparing it with standards interpreting differences, so that they conformed to standards.

SPECIFIC FACTORS OR VARIABLES TO BE CONTROLLED

1. **QUANTITY**-This is one of the variables that need to be controlled. The expectation normally is that specific output levels are expected from given quantities of inputs. A significant short fall is usually regarded as symptomatic of reliability problems, which can be trace to faulty operating systems leading to high levels of wastage. Alternatively in experience on the part of the operator can result in shortfall of output. The quality of raw materials used is a factor which may also explains why large wastage may result.

2. **QUALITY**-Organizations usually set quality standard for their products which ensures the attainment of specified performance levels for such products.

Quality deviations could occur due to the type of raw material used or the quality of manpower employed, significantly also quality differences may be explained by the process used.

3. **COSTS**- In the process of production there are a number of costs that are incurred which must also be controlled, Among them are:

- a. **Storage costs**; storage costs can be controlled by ensuring that an economic order quantity is held at any particular point in time.
- b. **Raw materials cost**: This can be controlled by buying materials at such points where prices are comparatively less and buying in economic lot size that attract quantity discount.
- c. **Ordering cost**: This can be controlled by ensuring that an economic order quantity is placed each time.
- d. **Cost of transportation**: This can also be controlled by adopting a transport model which brings minimum benefit to the organization.

THE ROLE OF EXTERNAL ENVIRONMENT IN PRODUCTION CONTROL

The external environment of an organization has certain influences on the organization, specifically the environment factors include:

- a. **CUSTOMERS**: the customers that patronize the product of an organization impose some external control on such an organization. This is done through a system of showing acceptance of the products by buying more or rejection of the product by withholding patronize of the product. When customers buy a firms product it is indicative of acceptance of the quality of the product. When patronage is withheld, customers show disapproval of quality levels. By these firms are forced to re-examine the quality of their products to re-work them to acceptance levels.

b. **COMPETITORS:** organizations face competition in the market place. These competitive forces encourage firms to produce at acceptable quality levels that can favourably withstand the product of rival firms and maintain ones market share.

c. **SUPPLIERS:** When suppliers of raw material bring in high quality raw materials, the firm will most likely produce high quality output. On the other hand if low quality raw materials are supplied the tendency is to produce low quality output.

d. **VALUE SYSTEM:** The value system of a society has to do with what the society hold as being sacred. Goods and services that individual firms or organizations produce are expected to conform to this value system. If they run contrary to the accepted norms of the society they are rejected.

e. **GOVERNMENT:** Government usually specify some regulation governing the production of certain goods. This is done to ensure that some minimum standards are attained for the production of some goods and services. An organization that intends to produce such goods and services must abide by such laws.

WORK STUDY

This is the systematic process of identifying the content of a job and measuring all aspect of the job in order to improve process for enhancement of efficiency and productivity.

Work study is decomposed in to two namely; Method study and work measurement.

PURPOSE OF WORK STUDY

The essence and purpose of work study include among others:

1. Establishing work standard that is aimed in ascertaining the basic requirements in labour, equipment, evaluating performance, planning activities, costing operations and products.
2. It establishes a framework for analysis, design and the improvement of work system, work stations, and methods of doing the work.
3. It determines the path and procedure leading to effective planning and control of work and material utilization.
4. It facilitates the specification of facilities, layouts, space utilization, material flow etc.
5. It facilitates design and application of effective job evaluation scheme and exercise.

METHOD STUDY

This is also known as method analysis. This is the systematic way of examining existing and new job in order to improve the sequence of work place and the movement of materials and the workforce for an enhanced productivity. The whole activities are aimed at developing strategies and mechanism that will improve the work process, which include workforce, material etc. in order to enhance performance at a reduced cost

OBJECTIVES OF WORK STUDY

The objective of work study is to assist management obtain the optimum use of the human and material resources available to an organization for the accomplishment of the work upon which it is engaged.

Fundamentally, this objective has three aspects.

1. The most effective use of plant and equipment.
2. The most effective use of human effort.
3. The evaluation of human work.

WORK MEASUREMENT

Slack, et al (1988) observed work measurement as the process of establishing time that a given task should take when performed by a qualified worker working at a defined level of performance. Stevenson, (1999) opines that work measurement is concerned with determining the length of time it should take to complete the job.

Work measurement is therefore a systematic application of techniques that is designed to establish standard time than an average worker would be capable of completing a task at a given level of performance.

WORK MEASUREMENT PROCEDURE

The following are the steps observed in the course of measuring work and they include:

1. Selection of task to be studied: the first course of measuring work involves the selection of the specific work or task to be studied from the work Centre or department of interest in the organization.
2. Recording facts about the task: After the selection of the task to be studied it is logically followed by a diligent exercise of sourcing for reliable and relevant facts about the task from the source document or the instrument adopted for tracking the information.
3. Analyzing the facts collected: the facts that are collected about the task are critically analyzed.
4. Measure the task: this is the particular step where the tasks that have been analyzed are measured in terms of its completion time based on the approach that has been adopted. The common approaches that are used include, time study, activity, sampling, analytical estimation predetermine standard time e.c.t
5. Compile the method: this is where the possible method that needs to be adopted are compiled and documented for onward decision.
6. Define the method and its standard time: After the methods are compiled, it is logical to define the method that will be adopted and the standard time for completing the task.

LABOUR STANDARD

Labour standard are used to establish work load, because the handling of the work load over any period of time dependent on the method used and the detailed analysis of that method. Labour standard can therefore be regarded as the determination of the time that an operation performed by a given method under given job conditions take when.

1. It is worked on by an operator who has the required skill, adequately trained to perform the operation properly.
2. The operator working on it is working at a pace that can be maintained throughout the day which is specified as equivalent to earn a best pay.
3. All of the operators needs are provided for. The time thus taken is as the standard time. There are a number of steps used to arrive at the standard time for any given operation.

These will include the following:

1. OBSERVED TIME: the observed time is the average time obtained in the experiment to complete one circle and the formula used is :

$$OT = \frac{\sum \mu_i}{n}$$

Where

OT = observed time

μ_i = the 1st observed time

n = number of observations.

Although observed time reflects actual data, it is not the measure that is actually used, because of some draw back associated with it. It is said that observed time does not take into account the operator. Therefore the observed time need to be modified in order to incorporate this element. This therefore gave rise to the concept of normal time.

2. Normal time: normal time adjust the observed time to take in to account the operators performance. This performance is measured by the performance rating (PR). A standard value of the performance rating

1.0 with a higher rating of 1.05 indicating that a faster than normal case is occurring and a lower rating 0.95 indicating the opposite.

The normal time is found by multiplying the observed time by the performance rating

NT = OT x PR Where

NT = normal time

OT = observed time

PR = performance rating

The normal time has also been criticized for not taking in to account an allowance factor to reflect delays, interruption that occurs during the normal work days. The delays may result from issues like:

Distortions due to equipment failure Schedule break.

Materials not being available on time as well as many other interruption to the work process it is therefore recommended that the normal time should be adjusted by an allowance factor (AF) to obtain a standard time.

3. To obtain a standard time from normal time we define the job percent allowance to all delays and interruption. This provides us with an allowance factor which is multiplied by the normal time to yield the standard time.

$$ST = NT \times AF$$

Where

ST = standard time

NT = normal time

AF = Allowance factor

To obtain the standard time directly from the observed time (OT) you can use the formula:

$$ST = PR \times AF \text{ Where}$$

ST = standard time

OT = observed time, PR = performance rating

AF = Allowance factor

USES OF LABOUR STANDARD

1. It provide data that are basic to many decision making problems in production/operation management. This involves cost of labour. This will enable the management to determine the exact amount to be allocated to labour. This is normally based on the quantity of work that will constitute a day's job.
2. It provides basic data used in day to day operation of the plant: under normal circumstance jobs should be scheduled to meet customer delivery date for instance a customer place an order for 30,000 tons of a product and the scheduled delivery date is 30th September from the 12th of July. This gives a period of 78 days where the company can produce 5 tons per day. We multiply the 5 tons x 78 days and divide by the 30,000 tons to obtain the period required to meet delivery promise.
$$= \frac{5 \times 78}{30,000}$$

3. It provides the basis for labour cost control: this means the data provided by the labour standard, can be used in measuring workers performances indexes.

INVENTORY MANAGEMENT

The management of inventory is very important in the life cycle of an organization because inventory is considered as the life wire of an organization. Lack of proper management of

inventory has every potential of disrupting the successful operations, reducing customers satisfactions, and intensifying operating costs. The effective management of inventory is aimed at striking a balance between the under stocking and overstocking of inventory. To achieve this balance is a matter of determining the quantity to order (optimum quantity) and when to order it (timing).

Therefore, inventory management is regarded as being important because inventory is a class to assets considered important to any manufacturing balance balances company. Large inventory balances require a significant investment of funds. This makes it necessary for most companies to establish policies for the management of inventory. A planned flow of materials is essential to the efficient operation of the manufacturing enterprise. There must be a synchronization of the input-output process. A manufacturing process will usually require to have a clear minimum of stock around which production schedule may be planned. Having too little stock (under stocking) of certain material can distort or hold production and therefore may adversely affect sales.

COST ASSOCIATED WITH INVENTORY

1. **CARRYING COST:** these are costs incurred in the process of storage or maintenance of inventory in the warehouse such costs includes security of the ware house, rentage of the ware house, insurance, lighting etc.
2. **ORDERING COST:** these are costs incurred in the process of placing an order and receiving the order up to the point of production. Such costs will include cost of paper work.
3. **OUTAGE COST:** This cost is as a result from the loss realized by the firm, if stock out of items occurs. Sales losses are experienced when inventories are not adequate to meet consumer demands. In some cases production stoppage occur if critical items of inventories are insufficient to meet needs

OBJECTIVES OF INVENTORY MANAGEMENT

There are several objectives for keeping inventories in organizations and they include:

1. To meet anticipated demand: basically inventories are kept at a measurable and sufficient level in order to meet anticipated demand of customers. It is against this reason that inventories are referred to as anticipation stocks.
2. To smooth and absorb variations in production and demand: organization that produce goods and services with seasonal pattern of demand always create room for building more inventories that will be able to smoothen and ensure that customers demands are sufficiently taken care of especially during peak period. The seasonal inventories include greeting cards, fresh fruits vegetables, rain coats, umbrella, ram during Sallah, etc.
3. To wedge against unnecessary stock outs: inventories are sufficiently maintained in order to wedge against possible stock outs that are likely to be experienced by organizations as a result of the undue delay and late deliveries of materials, unanticipated demand, quality problem and failure variation in weather and natural phenomenon, etc. keeping safety stocks serve as insurance against the relativity and variations in demand and lead time.
4. To tap synergy of bulk purchasing: usually organizations prefer to buy in bulk in order to enjoy some advantages that are associated with bulk purchasing. However, organizations resort to

periodic orders or order cycle by keeping cycle stocks so as to produce in economic lots sizes in as much as it is economical to produce on a large scale than small quantity.

5. To safe guard against price increase: when organizations foresee that there will be diametric increase in the prices of materials and other related inputs, they purchase large stocks of inventories in order to wedge and safe guard against successive increase in prices and to also enjoy the dividend of price discounts involved in large orders.

ECONOMIC ORDER QUANTITY (E.O.Q)

The E.O.Q is defined as the ordering quantity which minimizes the balance of cost between inventory holding costs and re-order costs over a given period as determined by individual companies.

ASSUMPTIONS OF E.O.Q

To be able to effectively calculate a basic E.O.Q model some critical assumptions become necessary:

- a. That the rate of demand is known to be continuous and continually represented by distribution that does not change in time
- b. That the replenishment of stock items is made instantaneously implying a supply lead time that is constant
- c. That there exists a known constant ordering cost.
- d. That there exists a known constant price per unit,
- e. No stock outs allowed

The E.O.Q model assumes that demand is constant, and that inventory is depleted at a fixed rate until it reaches zero. At that point, a specific number of items arrive to return the inventory to its beginning level. Since the model assumes instantaneous replenishment, there are no inventory shortages or associated costs. Therefore, the cost of inventory under the E.O.Q model involves a tradeoff between inventory holding costs (the cost of storage as well as the cost of tying up capital in inventory rather than investing it using it for other purposes) and order costs (any fees associated with placing orders, such as delivery charges). Ordering a large amount at one time will increase a small business, holding costs, while making more frequent orders of fewer items will reduce holding costs but increase order costs. The E.O.Q model finds the quantity that minimizes the sum of these costs.

In other terms the Economic Order Quantity (E.O.Q) is the number of units that a company should add to inventory with each order to minimize the total costs of inventory such as holding costs, order costs, and shortage costs.

The E.O.Q is used as part of continuous reviews of inventory system in which the level of inventory is monitored at all times and a fixed quantity is ordered each time the inventory level reaches a specific order point. The E.O.Q provide a models for calculating the appropriate re-order point and the optional re-order quantity to ensure the instantaneous replenishment of inventory with no shortages. It can be valuable tool for small business owners who need to make decision

about how much inventory to keep on hand, how many items to order each times, and how often to re-order to incurred the lowest possible costs.

Simply put, E.O.Q is the acronym for economic order quantity. The economic order quantity is the optimum quantity of goods to be purchased at one time in order to minimize the annual total costs of ordering and carrying or holding items in inventory.

The formula to calculate the economic order quantity is the square root of times the annual demand in units times the incremental cost to process an order) divided by (the incremental annual cost per unit to carry an item in inventory)

$$E.O.Q = x = \sqrt{\frac{2ADoc}{Hc}}$$

Example: Yewa and Sons Coca-Cola joint Federal Polytechnic Mubi has recorded an annual demand of 3,500 crates of soft drinks. A crate of soft drink cost N5, ordering costs are N30 per order and holding cost are charged at 25% of the costs of item. Calculate the economic order quantity.

$$E.O.Q = x = \sqrt{\frac{2ADoc}{Hc}}$$

$$AD = 3,500 \text{ Crates}$$

$$Hc = N5 \times 0.25 = N1.25$$

$$E.O.Q = x = \sqrt{\frac{2 \times 3,500 \times 30}{1.25}}$$

$$E.O.Q = x = \sqrt{165,000}$$

$$E.O.Q = 410 \text{ crates}$$

THE NEED FOR DATA COLLECTION STORAGE AND USAGE

Data collection and usage enhances use of inventory models. Some of the reasons for data collection and usage include the following:

- Updating the data collected at stores thereby ensuring constant information about inventories especially items issued out or supplied.
- It provides the management of a given organization the best solution regarding the use of available resources through those information to ascertain the objectives of the organization.
- It helps to avoid loss in sales by means of identifying or getting information from the data collected and stored.
- Auditing work would be easier, because requisite information has been collected and stored for further usage.
- Through data collection and storage stock levels can be effectively monitored thereby ensuring that unnecessary stock is eliminated.
- Data collected and storage are used for future purposes and also for reference purpose.

PURCHASING AND MATERIALS MANAGEMENT

Purchasing is an art, it not just getting something in change for money it is a profession and one at the basic factors that make up a business. Other factors that make up a business are finance, personal idea or design, manufacturing, distribution or marketing. The relative important of purchasing cuts across both areas. Efficiency is supplies are essentials to profitable trading. This is not to say that purchasing is as important in each and even industry as branch or branches at that industry. It is never the less, almost without exception the largest cost center present in all industrial business ventures in modern economy.

C.K Lysons also defined purchasing as the function responsibility for obtaining by purchasing, lease or o their legal means; equipment, materials supplies and services required by an understating for use in production or use in consumption.

Van Weele (2000) defined purchasing as the process of obtaining from external sources all goods and services, capability and knowledge which is necessary for running, maintain and managing the company primary and support activities at the most favorable conditions.

Generally, purchasing is a managerial process aimed at ensuring continuous flow of materials, projects and services in to and within the organization in order to meet it goals and objectives, profitability and efficiently in an ever changing business environment.

Materials management has been defined by the institution of materials management as the management process which integrates the flow of supplies in to, through and out of an organization to achieve a level of service which ensures that the right material are available at the right place, at the right time, of the right quantity and quality and at the right cost.

It include the function of procurement, material handling and storage, production and inventory control, packaging, transport and associated information systems, and their application throughout the supply manufacturing, services and distribution sectors.

PRINCIPLES OF MATERIAL MANAGEMENT

1. Planning: preparation of materials budget, product research and development, value engineering and analysis, standardization of specification.
2. Procurement: Determining order quantities, processing of work and stores requisition issues, of enquires, evaluation of quotations, supplier appraisal, negotiation, planning of contracts, processing of deliveries, certifying payment, vendor rating.
3. Storage: storage location, layout and equipment, mechanical handling, stores classification, coding and cataloging, receipt of purchased items, inspection, storage or return, protection, of stores issues to production, provision of cost data, stock record, disposal of absolute, or surplus scrap material.
4. Production control: forward ordering arrangement for material, preparation of production schedules and sequences, issues of orders to production, emergency action to meet shortages, make or buy decisions quality and reliability feedback and adjustment of supplies, flow to production line or sales trend.
5. Distribution: warehousing, packaging, external transportation.

INTERFACE OF PURCHASING WITH OTHER FUNCTIONS IN PRODUCTION

In any organizational set up achievement of the corporate goal is largely depends on the coordinated effort of the functional area managers. More especially the purchasing department by its very nature needs to have close and sustained relationship with many departments in the company as well as with the suppliers.

PURCHASING AND PRODUCTION

In discussing the relationship between purchasing department and production department, it is obvious that the principal service that the purchase department renders is the procurement of materials needed for production purposes. The production department relies on the purchase department giving them their specifications and training for material suppliers.

PURCHASING AND SALES

The sales department is responsible normally for marketing the finished products and the success of this exercise is very closely dependent upon the quality of the product, its price and the reliability of delivery dates. This therefore largely depends on the efficiency of the purchasing department and the production department among others. For example, the ultimate price charged for the product will, to a great extent depends upon the price which the purchase department has paid for the raw material and the efficiency of handling production possesses.

PURCHASING AND DESIGN ENGINEERING

The design or engineering department is saddled with the responsibility for designing the product to be manufactured. The purchase department helps to obtain quotations and technical assistance from suppliers and can help the design engineering department prepare and evaluate alternatives in regard to the specifications or design of the product or equipment. The purchasing department is also handling when the need to conduct market surveys and the need to search for new and better materials and equipment arises. The engineering department is expected to give the right specifications to the purchasing department.

PURCHASING AND MAINTENANCE

The maintenance department is expected to restore plants and equipment to their original functional state. This process will significantly rely on materials, spare parts and components supplied by the purchasing department.

PURCHASING AND PLANING

The planning department prepares long and short term plans for the operations of the organization. The plans are normally prepared in the content of environment in which the organization is operating. The purchase department being in constant touch with market will normally help the planning department draw up plans by keeping tabs in market development and the charges there in. it can also help the planning department by its forecast regarding the position of material supplies and market, particularly about the new products and new sources of supply.

The planning department should confide in the purchasing department keeping their in the full picture of the plans.

PURCHASING AND FINANCE

The finance department arrange for funds to meet suppliers obligation. It is expected that sufficient motive should be given to the finance department when funds are needed. The finance department will ensure compliance with financial regulations. All materials purchased must also be properly invoiced and receipted. Prices must also be in accordance with the purchase order. The finance department also ensures prompt payment to ensure a cordial relationship between suppliers and purchaser.

PURCHASING AND LEGAL

Purchase order is a legal contract between the buyer and the suppliers. The legal department provides expert interpretation of the law, contractual obligation commitment, agreement etc. the purchase department should always consult legal department when in doubt.

PURCHASING AND PERSONNEL

The efficient functioning of the purchasing department will depend upon the efficiency of the personnel department to carry out the function in respect of staffing.

PURCHASING AND STORES

A close relationship between purchasing department and stores department is necessary in order to ensure that materials are ordered strictly in term of actual requirement and in the overall interest of economy. The stores department receives and keeps adequate records of materials for easy retrieval and issuances to user units. The records maintained help to determine re-order levels and points in time when the purchasing unit should place order.

THE ROLE OF PURCHASING AND MATERIAL MANAGEMENT

The following are the role of purchasing and material management in an organization:

1. Procuring material: Procuring all the necessary materials needed for production or daily operation of the company or government organization. For a manufacturing company, this might include raw materials such as iron, steel, etc. but it might also include tools, machineries, delivery trucks or even office supplies needed for the secretaries and sales team. In retail environment, the purchasing department ensures that there is always sufficient product on the shelves or in the warehouse to keep the store well stocked. With a small business, it is especially important to keep inventory ordering at a reasonable level.
2. Evaluating price: the purchasing and material management department is charged with the responsibility of continuously evaluating whether it is receiving these materials at the best possible price in order to maximize profitability. This can be challenging for a small business that may purchase in lesser quantities than a larger vender and which thus may not receive the type of bulk discounts.
3. Paperwork and Accounting: Handling of all the paper work involved with purchasing and delivery of supplies and materials. Purchasing ensures timely delivery of materials generates and tracks orders and works alongside the receiving department and the accounts payable department to ensure that promised deliveries were received in full and are being paid for on time.

4. **Policy compliance:** Complying with all company's policies is very important. For example in a small business, before making a purchasing, the purchasing and material management must ensure that it needs the proper protocols for purchasing and budget approved and must ensure that any items purchased in accordance with the overall purchasing policy of the organization.
5. **Production and materials control:** production manager prepares schedules of production to be carried in future. The requirement of parts and materials are determined as per production schedules. Production schedules are prepared on the basis of order received or anticipated demand for goods. It ensured that every type or part of materials is made available so that production is carried on smoothly.
6. **Non-production stores:** Non-production materials like office suppliers, perishable tools and maintenance, repair and operating suppliers are maintained as per the needs of the business. There stores may not be required daily but their availability in stores is essential. The non-availability of such stores may lead to stoppage of work.
7. **Transportation:** the transport of materials from suppliers is an important function of material management and purchasing unit. The traffic department is responsible for arranging transportation services.
8. **Material Handling:** it is concerned with the movement of materials within a manufacturing establishment and the cost of handling materials is kept under control. It is seen that there are no wastages or losses of materials during their movement. Special equipment may be acquired for material handling.

FORECASTING AND PRODUCTION PLANNING

Forecasting has been defined as the use of a numerate and logical system, incorporating judgment values, to evaluate the probabilities of future occurrence or outcomes. Forecasting should not be regarded as a scientific method of prophecy where the future is known with certainty. Forecasting identifies possible outcomes and assigns them appropriate probabilities.

Forecasting is based on a logic of known relationship and past experience, additionally an assessment of how people or event are likely to react as judged from that experience, after allowing for known or probable changes that may occur in the said environment. Every forecast can be falsified.

The value of any forecast lies as much in its analysis, of the operative factors in relation to the environment and its indication of what likely changes, together with their limits, as it does in assessing the probabilities of the likely outcomes.

Generally it should be noted that a forecast cannot say what should be done, it can, of course test the options open to the decision taker, it cannot choose them.

In other words forecasting is the process of making predictions of the future based on past and present data and analysis of trend.

Uncertainty is regarded as the natural state of the world as it is known. It can not be totally eradicated. At best we can understand uncertainty and use mathematics or judgment or both to quantify it and that way convert it in to risk.

Risk is inevitably remains a personal assessment. Irrespective of the amount of sophistication attributable to the means used to measure the element of personal confidence in those means. Conceptually, this can be regarded as the equivalent of the statistician's confidence level-that's the inherent risk that has to be accepted. This therefore calls for dialogue between the specialist and the managers. Forecasting provides the opportunity required for such dialogue.

Risk and uncertainty are central to forecasting and prediction, it is generally considered good practice to indicate the degree of uncertainty attaching to forecasts. In any case, the data must be up to date in order for the forecast to be as accurate as possible.

Business forecasting helps directs and managers to get confirmation of the acceptance of their effort to meet the need of customers satisfactorily and at a profit.

PRODUCTION PLANNING AS IT RELATES TO FORECASTING

Production planning is concerned with the determination, acquisition and arrangement of all facilities necessary for future operations. Similarly, is a pre-determination of manufacturing requirements such as manpower, materials, machine and manufacturing processes.

It prepares a detailed plan for achieving the production goals economically, efficiently and in time. It forecasts each step in the production process. It forecasts the problems, which may arise in the production process. It tries to remove these problems. It also tries to remove the causes of wastage. Production planning is a tool available to management to achieve the stated objectives.

BASIC FORECASTING TECHNIQUES

Sales forecast is a prediction of expected sales by products and price for number of months or years. It represents a type of proforma sales portion of the traditional income statement for the future. When sales forecast is made over a period considered enough in which management has some confidence, a good job of forecasting profits and cash flow of the business can be done.

Below are the forecasting techniques:

SIMPLE TREND ANALYSIS

In this technique managers review historical data and use the rate or amount to change to future trends. The problem that goes with this technique is that it ignores environmental changes. For example, changes in consumer taste, economic conditions, competition, and a host of other factors may changes the sample trend. Thus although it is easy to use, simple trend analysis can mislead in that it assumes that the future will be like the past.

MARKET SHARE ANALYSIS

Market share analysis start with the assumption where in a particular firm has the view that its market share will remain constant. Then if the firm expects that the total industry sales grow by a given percentage, say, ten percent (10%) in the fourth coming year, it can expect its sales will grow by ten percent (10%) in that same year. For relatively stable industry this techniques will appear to work effectively. In a situation of rapid changes in the industry this assumption of market share remaining constant may be difficult to sustain.

MARKETING TEST

Sometimes no historical data are available on which to base projections. This is often the case with new productions. In such situations firms often conduct marketing test programmes in which the product is introduced in a few specific markets. When the marketing test has been completed, the firm then uses the results to measure the total sales for the region in which the product will be introduced.

The technique has some attendant problem. This is because economic, political, competitive and other environmental conditions are not necessarily the same from one location to the next. Again, competition often employs disruptive techniques, such as increasing their advertising to distort market test result. Equally, competitors investigate product brought out for test in the market and use that information to improve their own products.

THE MARKETING BUILD UP TECHNIQUE

Here firms use one of the other forecasting methods as a means of gathering information on the specific market segment and then aggregate them to arrive at a total sales forecast. Such segments may for example be based on geographic, age categories, ethnic groups or even product uses. For example a producer of egg may know that there are two basic type of users of it product that is people who use it for baking and those who use it for direct consumption. The firm may first estimate sales for each segment and then aggregate them.

JURY OF EXECUTIVE OPINION

Here executive from various unit and departments of an organization are asked to estimate market potential and sales for several years in to future. They in turn will try to arrive at a consensus. This method relies on the use of historical data, and experience and judgment as basic for decision making. The drawback of this system however is that some executives may be out of touch with the market and may overlook important factors.

SALES PEOPLE ESTIMATE

In this approach sales people are asked to estimate sales in their territories. This advantage of this approach remains that the sales people are assumed to be more aware of buyer's wants and need and their likely reactions to change in marketing mix.

STATISTICAL TECHNIQUES

In this approach historical data are collected or each variable and fitted in to a mathematical model. An example of such technique is regression analysis. With the aid of a computer an equation is developed that shows how each variable affect the events being forecasted. The analyst then "plugin" values for each variable to come up with a forecast.

QUANTITATIVE APPROACH

In this approach we use figures as the basis for calculations. The approach is based on the assumption that past history is indicative of future expectations and past data is used as the basis for making projections. A good example here is the trend extrapolating, which involves picking a tracking factor or environmental variable and taking note of its trend statically. The usual practice is to fit a line to historical data and extending the line in to future period.

The regression model is another example of the quantitative approach to forecasting. This functionally relates sales to other economic variables using the estimates of the data. Such data must be some easy method for developing a forecast from past data are the simple moving average, weighed moving average, exponential smoothing and regression analysis.

Since production planning is concerned with developing a specific course of action for the production system over a given period of time, it therefore becomes expedient to have a good forecast for the level of demand for the firms product line which will aid the selection of the combination of human and material resources that can produce the necessary output to meet that level of demand efficiently without a good forecasting exercise production planning will be a meaningless exercise.

FACTORS AFFECTING THE RELIABILITY OF FORECAST

Measurement of forecast must have a consistent base. There is also need for probability to be attached thereby evaluating the likelihood of events occurring.

1. The data used: this will normally affect the reliability of the forecast to achieving its purpose. The use of poor data results in significant deviation between our forecast and what actually obtains.
2. The span of forecast: Generally, the shorter the time the more reliable is the forecast other things being equal. A forecast with long span of time is usually less reliable due to the increase chance of occurrence of other events that will nullify the forecast.

CONCEPTUAL ISSUES IN FORECASTING

Stevenson, (2005) opines that forecasting is a statement about the future value of a variable of interest. It is act of determining the future value of activities by using some predefined set of data and variables in order to make a logical judgment and decision.

A planning tool that helps management in its attempts to cope with the uncertainty of the future, relying mainly on data from the past and present and analysis of trends.

Forecasting starts with certain assumptions based on the management's experience, knowledge, and judgment. These estimates are projected into the coming months or years using one or more techniques such as Box-Jenkins models, Delphi method, exponential smoothing, moving averages, regression analysis, and trend projection. Since any error in the assumptions will result in a similar or magnified error in forecasting, the technique of sensitivity analysis is used which assigns a range of values to the uncertain factors (variables). A forecast should not be confused with a budget.

Estimate future trends by examining and analyzing available information.

It is the use of a numerate and logical system, incorporating judgment, values to evaluate the probabilities of future occurrences or outcomes.

FEATURES COMMON TO ALL FORECASTS

A wide variety of forecasting techniques are in use. In many respects, they are quite different from each other, as you shall soon discover. Nonetheless, certain features are common to all, and it is important to recognize them.

1. Forecasting techniques generally assume that the same underlying causal system that existed in the past will continue to exist in the future. Thus, a manager cannot simply delegate forecasting to models or computers and then forget about it, because unplanned occurrences can wreak havoc with forecasts. For instance, weather-related events, tax increases or decreases, and changes in features or prices of competing products or services can have a major impact on demand. Consequently, a manager must be alert to such occurrences and be ready to override forecasts, which assume a stable causal system.
2. Forecasts are rarely perfect; actual results usually differ from predicted values. No one can predict precisely how an often large number of related factors will impinge upon the variable in question; this, and the presence of randomness, precludes a perfect forecast. Allowances should be made for inaccuracies.
3. Forecasts for groups of items tend to be more accurate than forecasts for individual items because forecasting errors among items in a group usually have a canceling effect. Opportunities for grouping may arise if parts or raw materials are used for multiple products or if a product or service is demanded by a number of independent sources.
4. Forecast accuracy decreases as the time period covered by the forecast that is the time horizon increases. Generally speaking, short-range forecasts must contend with fewer uncertainties than longer-range forecasts, so they tend to be more accurate.

An important consequence of the last point is that flexible business organizations—those that can respond quickly to changes in demand—require a shorter forecasting horizon and, hence, benefit from more accurate short-range forecasts than competitors who are less flexible and who must therefore use longer forecast horizons.

QUALITIES OF A GOOD FORECAST

The bottom line of every forecasting is to achieve a satisfactory level of result with minimum degree of errors, variance between the actual and forecast values. Thus, forecast is said to be good if it posess the following qualities;

1. Reliability: A good forecast is expected to be reliable by working consistently and producing a satisfactory result at all times.
2. The forecast should be timely. Usually, a certain amount of time is needed to respond to the information contained in a forecast. For example, capacity cannot be expanded overnight, nor can inventory levels be changed immediately. Hence, the forecasting horizon must cover the time necessary to implement possible changes. A good forecast is always expected to be defined within a reasonable time frame that will guarantee proper implementation and adjustments of possible changes. The time frame should be spelt out to allow adequate transformation of data and information meant for the forecast.
3. The forecasting technique should be simple to understand and use. Users often lack confidence in forecasts based on sophisticated techniques; they do not understand either the circumstances in which the techniques are appropriate or the limitations of the techniques. Misuse of techniques is an obvious consequence. Not surprisingly, fairly crude forecasting techniques enjoy widespread popularity because users are more comfortable working with them.

4. The forecast should be in writing. Although this will not guarantee that all concerned are using the same information, it will at least increase the likelihood of it. In addition, a written forecast will permit an objective basis for evaluating the forecast once actual results are in.
5. The forecast should be accurate and the degree of accuracy should be stated. This will enable users to plan for possible errors and will provide a basis for comparing alternative forecasts.
6. Cost-effective: Cost is an essential element for judging the effectiveness and quality of a forecast. When the cost of a forecast supersedes its relative benefits, the forecast is regarded to be bad and good when its benefits supersede its relative costs.
7. Ability to be expressed in meaningful terms a good forecast should always be expressed in meaningful terms or units for easy understanding and application in any technique. When this quality is not well specified, it will affect the validity and reliability of the forecast. For instance, the terms can either be defined in units or, monetary values base on the need and choice of the users.
8. Forecast should be in writing: A good forecast should be given in a written format in order to lend itself to an objective ground for evaluating the variability of errors, once the actual results are obtained.

STEPS IN FORECASTING

Forecasting involves the following steps;

- i. Determining the intention of the forecast.
- ii. Establishing a time frame (horizon).
- iii. Selecting a forecasting technique.
- iv. Generate and analyse relevant data.
- v. Conduct the forecast.
- vi. Monitor and control the forecast.

BENEFITS OF FORECASTING

The benefits of forecasting include;

- It ensures that production planning activities are determined on the basis of realistic and dependable data.
- It serves as a base that directs the affairs and activities of production managers.
- It ensures that firms always maintain a balance stock level in order to avoid the consequences of over-stocking or under-stocking.
- It tends to reduce operations cost by budgeting for resources only meant for immediate utilisation.

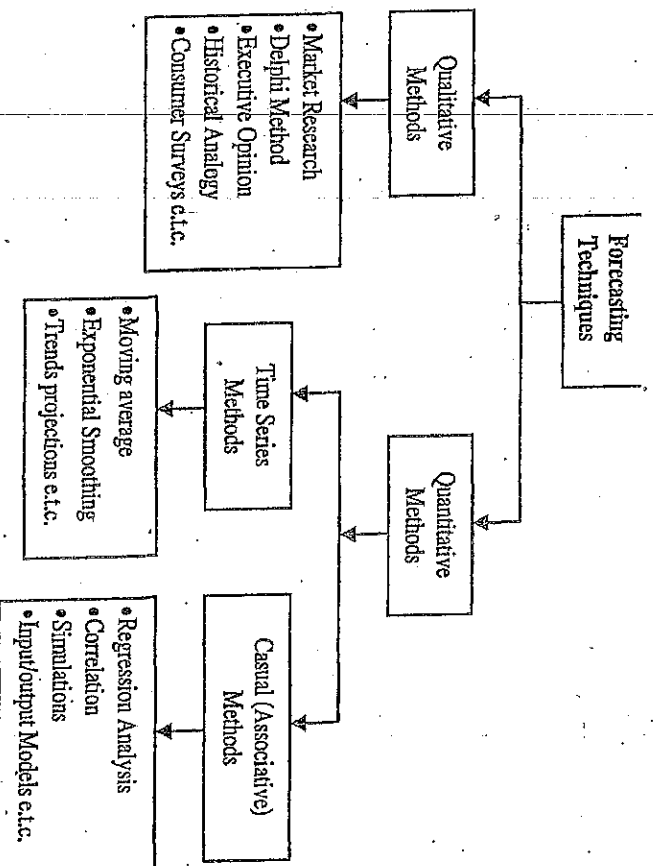
SHORT-COMINGS OF FORECASTING

Despite, the benefits of forecasting, it is also identified with short comings which include;

- It limits individual dynamism and flexibility in as much activity is based on budgets.
- It consumes more time and is not easy to come by.
- It is costly in nature (i.e. the cost of running the forecast system its validity and reliability is prone to uncertainties in environment.
- It needs highly skilled and motivated workers to maintain the system.

FORECASTING METHODS

There are two methods by which forecasting values are obtained the methods include; the qualitative and quantitative methods



QUALITY CONTROL IN PRODUCTION

Quality control can be defined as the means of tracking the quality characteristics of a product. Those quality control operations are normally delegated to the quality control department. It is their job to monitor the quality characteristics of the firm's product.

Quality control (QC) is also a procedure or set of procedures intended to ensure that a manufactured product or performed service adheres to a defined set of quality criteria or meets the requirements of the client or customer.

Similarly is the process of inspecting products to ensure that they meet the required quality standards.

To a manufacturer, quality is concerned with making products or providing services that are free of errors and that conform precisely to their design specification.

To a user, quality may mean the product or service fits for its purposes. In sum, quality can be defined as the degree of fit between customer's expectations and customer perception of the product or services. Quality is consistent with performance to customer's expectations.

Additionally, quality control requires the business to create an environment in which both management and employees strive for perfection. This is done by training personnel, creating benchmark marks for product quality, and testing products to checks for statistically significant variations. Finally, the quality control processes that must be ongoing to ensure that remedial

efforts, if required, have produced satisfactory results and to immediately detect recurrences or new instances of trouble.

The operations of the quality control scheme usually require the setting of standard by an organization. These standards become the bench mark against which comparisons can be made. The standard normally specified performance level associated with the particular product. Organizations therefore, strive to ensure that the product manufactured does not perform.

METHODS OF QUALITY CONTROL

Below are some of the methods of quality control in an organization: inspection, sampling and control chart.

INSPECTION : this is an appraisal activity that compares goods and services standard. Inspection can occur at three point: before production i.e, at the receiving point of raw materials, during production i.e, work-in-process or conversion and after production i.e, output inspection deals with this basic issues; how much to inspect and how often, at what point in the process should inspection occur, whether to inspect in a centralized or onsite location and whether to inspect attributes.

SAMPLING PLAN: is a blueprint for how a sampling event or program will be executed. It also a detailed outline of which measurements will be taken at what times, on which material, in what manner, and by whom. It should provide all the detail needed to ensure that representative samples are collected, handled, analyzed, and reported in a manner that meets the needs and objectives of the sampler. Implementing a clearly defined and consistently employed sampling protocol reduces the chances of error that the sampling process.

OC Curve: is a graph used in quality control to determine the probability of accepting production lots when using different sampling schemes. It shows

Percentage defectives along the horizontal (X), axis and probability of acceptance along the vertical (Y) axis. Lots having more than the acceptable percentage of defectives are rejected. Operating characteristic (OC) curves are powerful tools in the field of quality control, as they display the discriminatory power of a sampling plan.

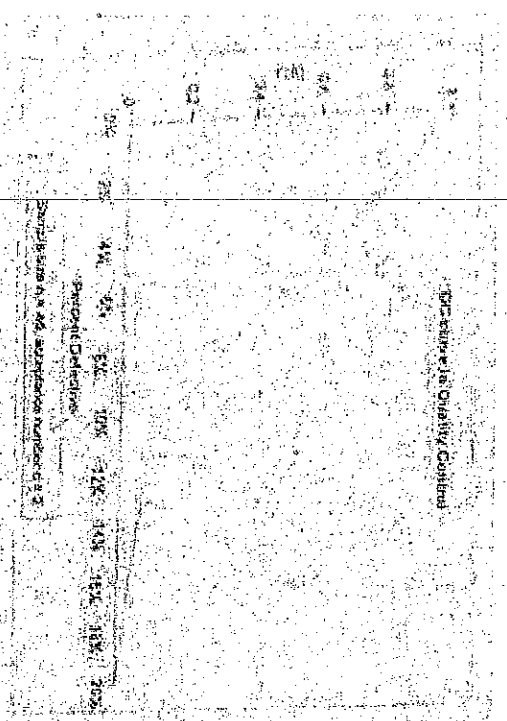


Figure 1: OC Curves in Quality Control

F-Chart : is a clear graphic representation of a process from beginning to end regardless of whether that process is an algorithm or manufacturing procedure. The tool once used exclusively by computer programmers has now become a fundamental quality tool. Flowcharting is typically used to map a process flow showing the beginning of a process, decision points and the end of the process. If company had to choose only one quality software, it should be a flow chart of some kind.

TRANSPORTATION IN PRODUCTION

The transportation model was developed independently in the 1930 and 1940s by three different researchers Lenoir Kantorowict, Tjallingis Koopmas and Frank Hitchcock.

The transportation model can be regarded as a special class of the linear programming problem. The model explains transportation methods which involves a number of shipping sources such as factories to a number of destinations such as warehouses. At every given time period each shipping sources (factory) is assumed has a certain capacity requirement and there is a given cost of shipping from the designated source to the required destination. The ultimate objectives is to determine the amount shipped from each source to each destination that minimize the total shipping cost while satisfying both the supply limits and the demand requirement.

In using the transportation method procedure, the sum of the supply sources must equal the sum of the demand destinations. We must in addition know with certainty the available s_i supply (at each of the supply source), the required d_j demand (at each of the demand destination), and the unit cost to ship a unit from each supply source to each demand destination.

To solve a transportation problem one can use one of the most common computer based method which is called the modified distribution method (MODI) which is an algebraic process of allocating the supply and demand requirement normally based on an index calculated from the cost values in each cell of the transportation problem. A computer software now takes care of the rather tedious process of calculation manually.

Several other procedure include an initial solution procedure, for test degeneration, a test for optimality and cell relocation procedure.

RESOURCE ALLOCATION

Resource allocation is the assignment of available resources to various uses in the context of an entire economy, resources can be allocated by various means, such as market or central planning. In project management, resources allocation or resource management is the scheduling of activities and the resources required by those activities while taking in to consideration both the resources availability and the project time.

PROJECT MANAGEMENT

Before understanding what project management is all about it is logical to start with the definition of a project.

A project is the combination of interrelated activities and tasks that are designed in order to produce an output at one time operation within a given time frame. The following activities are regarded as projects, designing new products or services, designing advertising campaign, designing information system etc.

Project management is the planning and controlling of all interrelated activities and resources that are committed in order to sustain a project in an

Programming: this refers to the solution, method which is normally carried out by an interactive process whereby one moves from one solution to a better solution progressively on until he reach a solution which cannot be improved upon and therefore, regarded as the optimum solution. The term programming makes use of some mathematical techniques in order to arrive at the best solution. i.e optimum solution utilizing the firm's limited resources.

Programming is also sometimes referred to as computing which is defined as calculating some unknown from a set of equations and or inequality under certain conditions expressed mathematically.